CRICKSTATS - Cricket Player Performance Tracking Application

Complete Project Documentation with Screenshot Guidelines

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User Story

As a Salesforce Administrator, I aim to design and implement a comprehensive Player Performance Tracking Application within Salesforce to record and analyze detailed, match-by-match performance data of cricket players. The application will utilize custom objects such as Player and Match Performance, each containing well-defined fields and relationships to ensure accurate data management and maintain data integrity. By implementing Master-Detail and Lookup relationships, the system will establish strong connections between players, matches, and performance records. This structure will enable efficient tracking of both individual and team performance metrics, support the creation of meaningful reports and dashboards, and allow stakeholders to analyze player statistics effectively. Ultimately, the application will improve data visibility, strategic decision-making, and player development processes through reliable performance insights.

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1. INTRODUCTION

1.1 Project Overview

Hey! Let me explain what CRICKSTATS does. Imagine you're coaching a cricket team and you need to track every player's performance—runs scored, wickets taken, catches, everything! Normally, you'd use spreadsheets or notebooks, and it gets messy fast. CRICKSTATS brings all that data into one organized Salesforce application!

What does it do?

CRICKSTATS is a Salesforce-based application developed to track, manage, and analyze the performance of cricket players on a match-by-match basis. The system leverages Salesforce features such as custom objects, object relationships, dashboards, and automation tools to organize and present player performance data in a structured manner.

Who uses it?

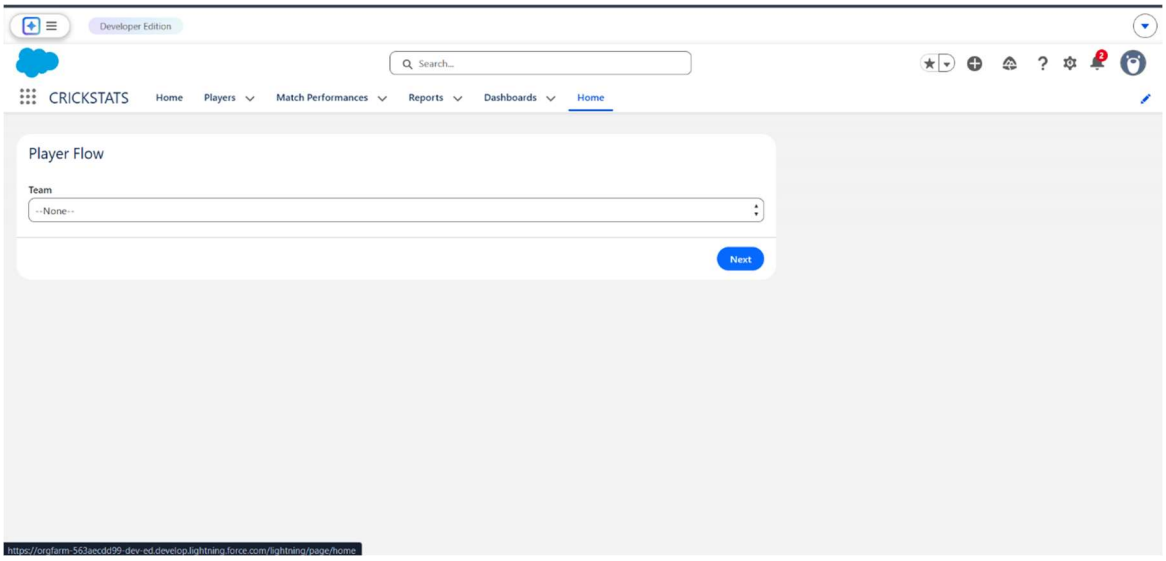
- **Coaches** - Track player statistics and make team selection decisions
- **Analysts** - Generate performance reports and identify trends
- **Administrators** - Manage player records and match data
- **Team Managers** - Monitor overall team performance

The cool automation features:

- Automatic calculation of batting averages, strike rates
- Roll-up summaries that count total runs, wickets, catches automatically
- Interactive dashboards showing player statistics by team
- Match performance records linked to players
- Real-time performance analytics

Through interactive dashboards and analytical tools, CRICKSTATS helps stakeholders gain meaningful insights into player performance, enabling data-driven strategic decisions and effective player development.

[SCREENSHOT : Project Overview Dashboard]



1.2 Purpose

The main purpose of this project is to develop an efficient system for managing and analyzing cricket player performance using the Salesforce platform. The project aims to:

- **Digitize and centralize player performance records** - Replace manual scorebooks and scattered spreadsheets
- **Improve accuracy and accessibility** - All cricket statistics in one structured database
- **Provide real-time analytics** - See player performance instantly through reports and dashboards
- **Support player development** - Help coaches evaluate performance trends and make better decisions
- **Automate data management** - Use Salesforce features like custom objects, relationships, workflows, and dashboards

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2. IDEATION PHASE

2.1 Problem Statement

Let me tell you what problems we're solving here.

Customer Problem Statement Template:

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-1	Cricket team coach	Track player performance across matches	I cannot get accurate statistics easily	Data is stored in different files and formats	Frustrated and unsure about team selection
PS-2	Team manager	Maintain complete player records	It takes too much time to update and verify data	Most records are maintained manually	Stressed and inefficient
PS-3	Sports analyst	Analyze player trends and generate reports	I face difficulty in comparing performances	There is no centralized analytics system	Confused and dissatisfied

Table 1: Problem statements from users

Before this system:

- Coaches were tracking player stats in notebooks and Excel files

- Team managers spent hours updating records manually
- Analysts couldn't easily compare player performances across matches
- No centralized place to see all player statistics

CRICKSTATS fixes all that!

2.2 Empathy Map Canvas

Target Users: Team Managers, Coaches, Analysts, Salesforce Admins

We talked to cricket coaches and analysts and here's what they told us:

Think & Feel:

- Need accurate data for team selection
- Want better insights into player form

Hear:

- Feedback from management: "Who should we pick for next match?"
- Complaints: "Where are the latest statistics?"

See:

- Multiple data sources (scorebooks, apps, websites)
- Scattered performance records

Say & Do:

- Collect statistics manually after each match
- Spend time consolidating data from different sources

Pain:

- Data errors and inconsistencies
- Delays in getting performance reports
- Difficulty comparing players

Gain (What they want):

- Faster decision-making with real-time data
- Automation of statistics calculation
- Clear visual dashboards

2.3 Brainstorming & Idea Prioritization

Brainstormed Ideas:

We came up with different solutions:

- Web-based player management system (needs development)
- Mobile statistics tracker (limited features)
- Salesforce-integrated CRM (customizable and powerful)
- AI-based performance prediction (complex to implement)
- Dashboard-based analytics (good but needs platform)

Selected Idea:

Salesforce-based performance tracking platform (CRICKSTATS) ← We chose this!

Why Salesforce?

- Built-in database with custom objects
- Powerful relationship management
- Automatic calculations with roll-up summaries
- Interactive dashboards and reports
- Secure cloud storage
- Scalable for multiple teams and leagues

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3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Let me walk you through how the system is used:

Journey Stages:

1. **Login** - User logs into Salesforce CRICKSTATS app
2. **Enter Player Data** - Admin creates player profiles with details
3. **Record Match Performance** - After match, enter runs, wickets, catches for each player
4. **Generate Reports** - Create reports filtering by team, player, or date range
5. **Analyze Results** - View dashboards showing batting averages, bowling stats
6. **Decision Making** - Coach uses insights for team selection and strategy

Every step is tracked in Salesforce with automatic calculations!

3.2 Solution Requirements - Functional Requirements

Here's what the system actually does:

FR No	Requirement
FR-1	Player Registration - Add players with profiles
FR-2	Match Performance Entry - Record match statistics
FR-3	Dashboard Generation - Visual performance analytics
FR-4	Report Creation - Detailed statistical reports
FR-5	Flow Automation - Auto-update player stats

Table 2: Functional requirements

Non-Functional Requirements

These are the quality standards the system must meet:

NFR-1: Usability

The system should be easy to understand and use by coaches, managers, analysts, and admins without requiring technical training.

Requirements:

- Simple and intuitive user interface
- Clear navigation menus
- Easy-to-read dashboards
- Minimal learning curve

Examples:
A new coach should be able to:

- Log in
- Enter match data
- View player reports

...within a few minutes without special training. The Salesforce Lightning interface provides drag-and-drop dashboards and user-friendly forms.

NFR-2: Security

The system must protect sensitive player data from unauthorized access.

Requirements:

- Role-based access control
- Secure authentication
- Encrypted data storage
- Regular backups

Examples:

- Coaches can view player statistics
- Managers can edit records
- Analysts can generate reports
- Only admins can modify system settings

Salesforce Profiles and Permission Sets restrict access to confidential information.

NFR-3: Reliability

The system should function correctly without failure and store data accurately.

Requirements:

- No data loss
- Automatic backups
- Error handling
- System stability

Example: If a manager enters match statistics and loses internet connection, the data should be saved automatically when reconnected. Salesforce cloud backup ensures recovery in case of system failure.

NFR-4: Performance

The system should respond quickly and handle large volumes of data efficiently.

Requirements:

- Fast page loading
- Quick report generation
- Smooth dashboard rendering
- Minimal response time

Example: When an analyst generates a performance report for 100 players, the report should load within 2-3 seconds. Optimized Salesforce queries and caching improve response speed.

NFR-5: Availability

The system should be accessible whenever users need it.

Requirements:

- 24/7 system availability
- Minimal downtime
- Cloud-based access
- Disaster recovery

Example: Coaches should be able to access CRICKSTATS even during tournaments at night or on weekends. Salesforce provides high uptime with cloud hosting.

NFR-6: Scalability

The system should support increasing numbers of users, players, and records.

Requirements:

- Handle growing datasets
- Support more teams and leagues
- Expand storage capacity
- Maintain performance under load

Example: If CRICKSTATS is extended from one team to 50 teams, the system should still perform smoothly. Salesforce cloud infrastructure automatically scales storage and processing.

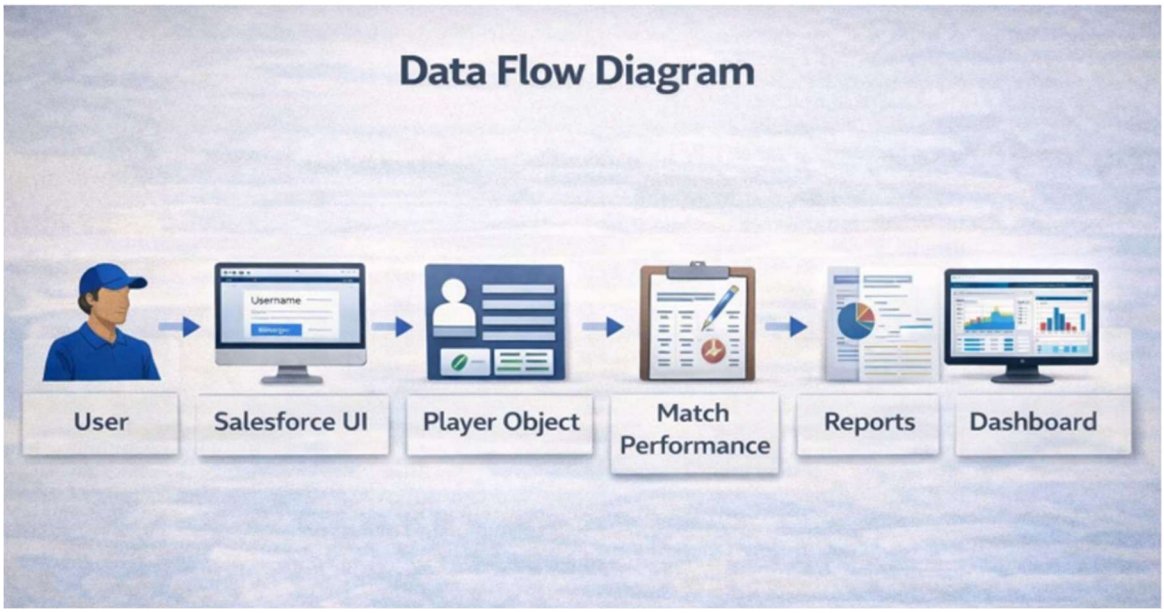
3.3 Data Flow Diagram

Here's how data flows through the system:

1. User enters data through Salesforce UI
2. Data stored in Player Object and Match Performance Object
3. Roll-up summaries automatically calculate totals (runs, wickets, catches)
4. Reports query objects and display statistics
5. Dashboards visualize performance metrics

The system processes match data, validates entries, calculates statistics, and generates reports—all automatically!

[SCREENSHOT : Data Flow Diagram]



3.4 Technology Stack

Here's what we're using to build this:

Layer	Technology
Frontend	Salesforce Lightning
Analytics	Reports & Dashboards
Automation	Salesforce Flows
Backend	Salesforce Objects
Database	Salesforce Cloud

Table 3: Technology stack

Everything is built within Salesforce—no external tools needed!

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4. PROJECT DESIGN PHASE

4.1 Problem-Solution Fit

The CRICKSTATS system effectively addresses the major challenges associated with traditional cricket performance management.

How we solve each problem:

Problem: Manual data management is time-consuming and error-prone
Solution: Automated system using Salesforce with real-time data entry and validation

Problem: Data loss due to improper storage
Solution: Secure cloud storage with automatic backups

Problem: Difficulty analyzing scattered data
Solution: Interactive dashboards with clear visual reports and insights

Problem: Delays in accessing performance information
Solution: Real-time reporting features for instant access to statistics

Problem: Can't compare player performances easily
Solution: Comprehensive reports with filters, groupings, and summaries

Overall, CRICKSTATS bridges the gap between traditional challenges and modern technological solutions!

4.2 Proposed Solution

CRICKSTATS provides:

- Custom objects: Player, Match Performance
- Master-detail relationships between objects
- Roll-up summaries for automatic calculations
- Lightning App for easy navigation
- Automated flows for data processing
- Interactive dashboards for visualization

Key Features:

- Centralized database - All player data in one place
- Live statistics - Real-time performance tracking
- Secure access - Role-based permissions
- Scalable design - Supports multiple teams and leagues
- Easy reporting - Generate reports in seconds

4.3 Solution Architecture

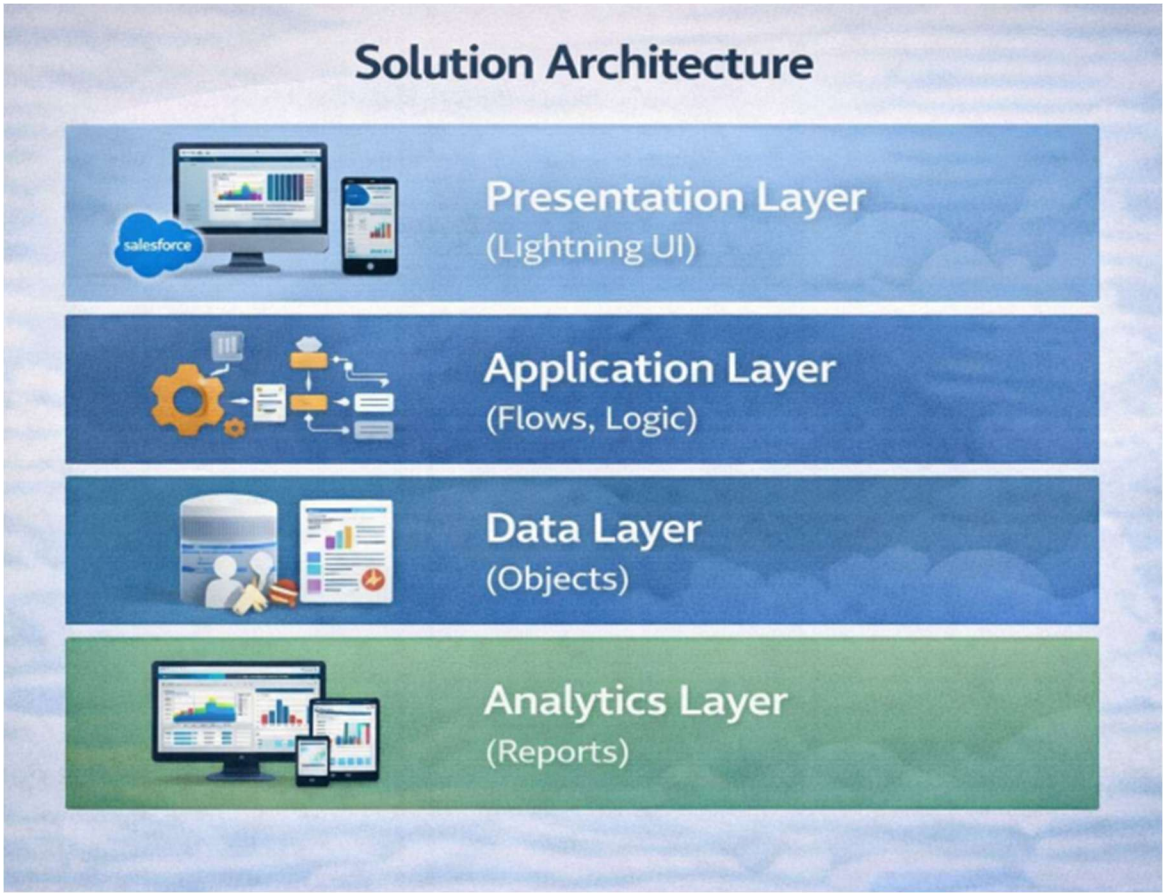
Architecture Layers:

- **Presentation Layer** - Lightning UI with custom app and tabs
- **Application Layer** - Flows and business logic
- **Data Layer** - Custom objects and relationships

- **Analytics Layer** - Reports and dashboards

Everything works together seamlessly in Salesforce!

[SCREENSHOT : Solution Architecture Diagram]



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5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

We used Agile methodology to build this. Here's how we broke down the work:

Sprint-1: Developer Setup & Object Creation

- USN-1: Create Salesforce developer account (PRITHIVIRAJ L) - 1 point
- USN-2: Create Player and Match Performance objects (ABISHEK) - 3 points

Sprint-2: Navigation, UI Design, Data Structure

- USN-3: Create separate tabs for each object (RAJA I) - 3 points
- USN-4: Create dedicated Lightning App (HARIKARAN E) - 3 points

- USN-5: Create player and match fields (PRITHIVIRAJ L) - 5 points
- USN-6: Customize page layouts (ABISHEK) - 5 points
- USN-7: Validate player and match inputs (HARIKARAN E) - 3 points

Sprint-3: Automation

- USN-8: Automated flows to update player statistics (PRITHIVIRAJ L) - 5 points

Sprint	Story Points	Duration	Start Date	End Date	Release Date
Sprint-1	20	1 Day	2 Mar 2026	3 Mar 2026	7 Mar 2026
Sprint-2	20	1 Day	3 Mar 2026	4 Mar 2026	7 Mar 2026
Sprint-3	20	1 Day	4 Mar 2026	5 Mar 2026	7 Mar 2026
Sprint-4	20	1 Day	5 Mar 2026	7 Mar 2026	7 Mar 2026

Table 4: Sprint schedule

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6. PROJECT DEVELOPMENT PHASE

Project Flow - Milestones

Let me walk you through each milestone step-by-step:

1. Creation of developer account
2. Object creation (Player, Match Performance)
3. Tabs creation
4. The Lightning App
5. Fields and Relationships
6. Roll-up Summary fields
7. Flows implementation

MILESTONE 1: Salesforce Developer Account Creation

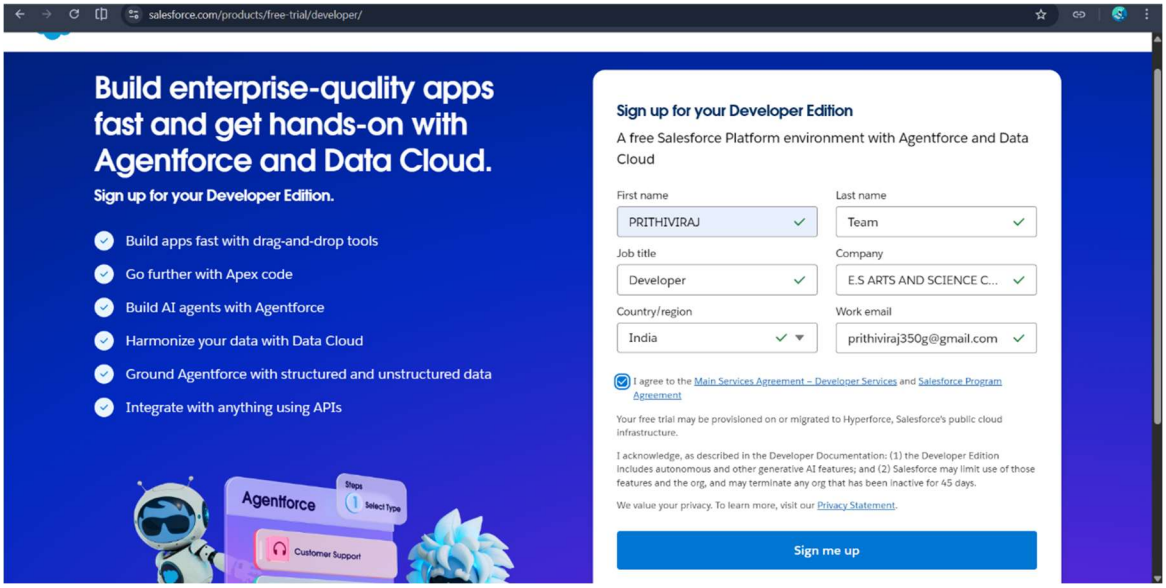
Activity 1: Creating Developer Account

Here's how to create your free Salesforce developer account:

Steps:

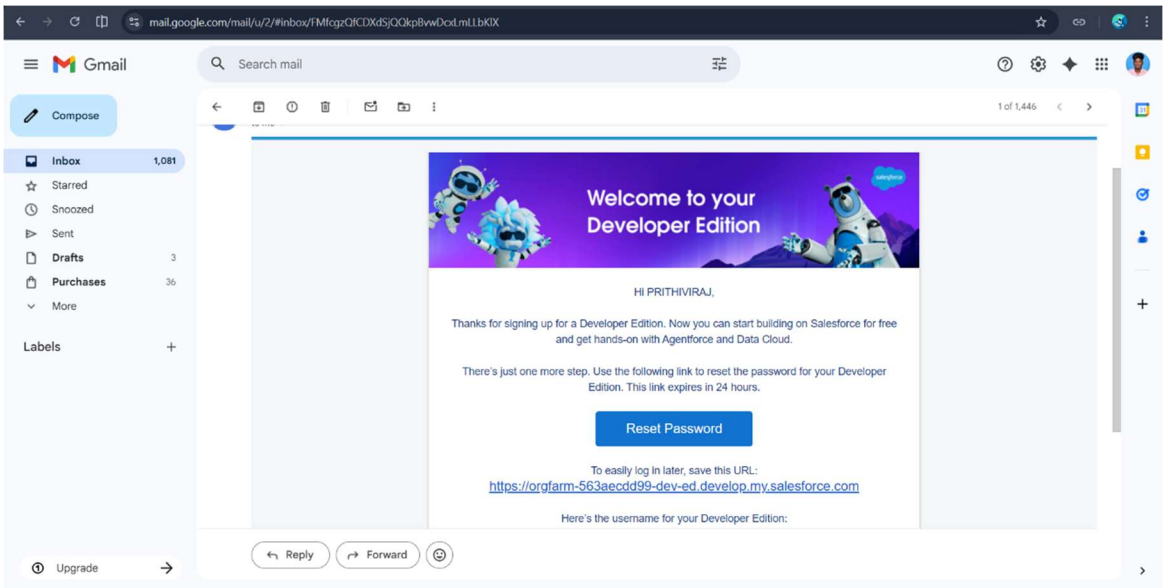
1. Go to https://developer.salesforce.com/signup
2. Fill out the signup form:
    - First name: Your first name
    - Last name: PRITHIVIRAJ Team
    - Email: prithiviraj350g@gmail.com
    - Role: Developer
    - Company: E.S ARTS AND SCIENCE CO-ED COLLEGE
    - Country: India
    - Username: prithiviraj350g@esarts.com (doesn't need to be a real email—just needs to be in email format)
3. Click "Sign me up"
4. Wait 5-10 minutes for the verification email

[SCREENSHOT : Developer Signup Form]



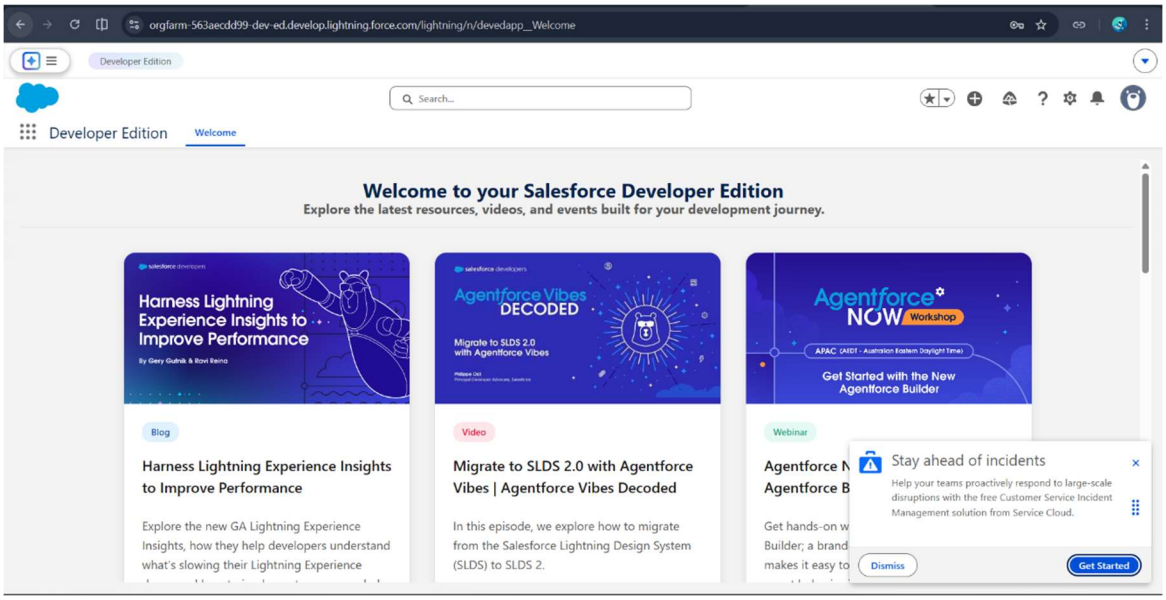
Activity 2: Account Activation

1. Check your inbox (prithiviraj350g@gmail.com) for the email



2. Click "Reset Password" in the email (activates your account)
3. Set a strong password
4. Answer a security question (pick something you'll remember!)
5. Click "Change Password"
6. You'll be redirected to your Salesforce setup page

[SCREENSHOT : Salesforce Welcome/Setup Page]



MILESTONE 2: Object Creation

Understanding Objects

Think of objects like database tables. Each object stores a specific type of information:

- **Player object** stores player information (name, team, role, batting style, etc.)
- **Match Performance object** stores match statistics (runs scored, wickets, catches for each match)

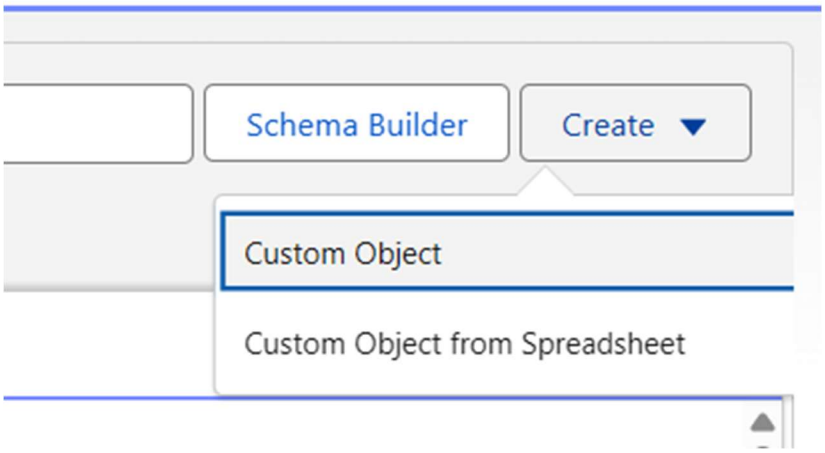
Activity 1: Player Object

Let me show you how to create the Player custom object.

Steps:

1. Go to Setup (click the gear icon, top right)
2. In Quick Find, type "Object Manager"
3. Click "Object Manager"
4. Click "Create" → "Custom Object"
5. Fill in the details:
    - Label: Player
    - Plural Label: Players
    - Record Name Label: Player Name
    - Data Type: Text
6. Under "Optional Features", check:
    - Allow Reports ✓
    - Allow Activities ✓
    - Track Field History ✓
7. Under "Search Status", check:
    - Allow Search ✓
8. Click "Save"

[SCREENSHOT : Player Object Creation Form]



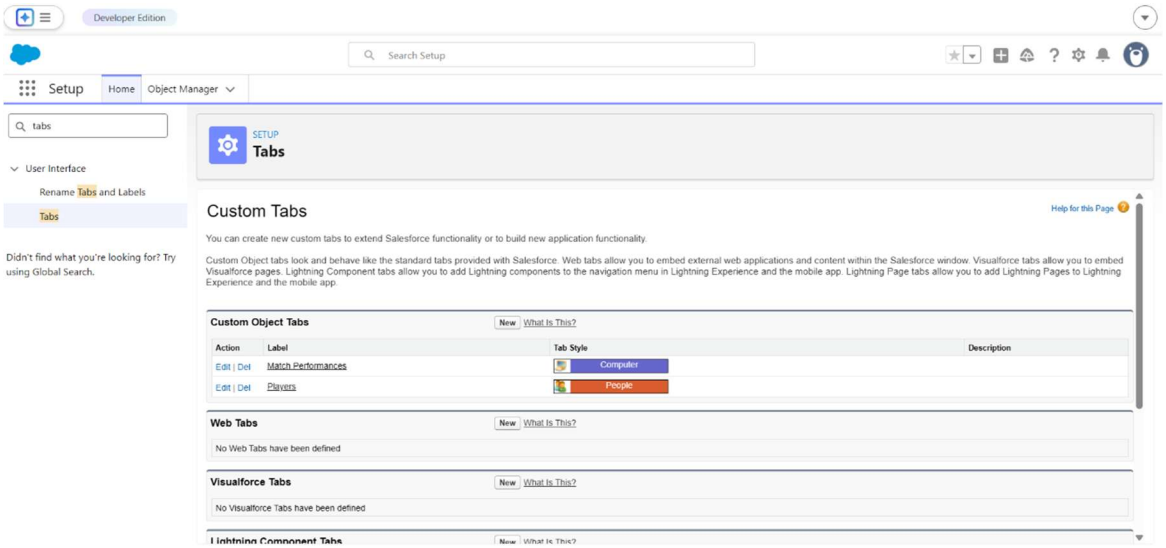
Activity 2: Match Performance Object

Now create the Match Performance object the same way:

Steps:

1. Go to Setup → Object Manager
2. Click "Create" → "Custom Object"
3. Fill in:
    - Label: Match Performance
    - Plural Label: Match Performances
    - Record Name Label: Match Performance Name
    - Data Type: Text
4. Check same Optional Features:
    - Allow Reports ✓
    - Allow Activities ✓
    - Track Field History ✓
5. Check "Allow Search" ✓
6. Click "Save"

[SCREENSHOT : Both Objects in Object Manager]



MILESTONE 3: Tabs

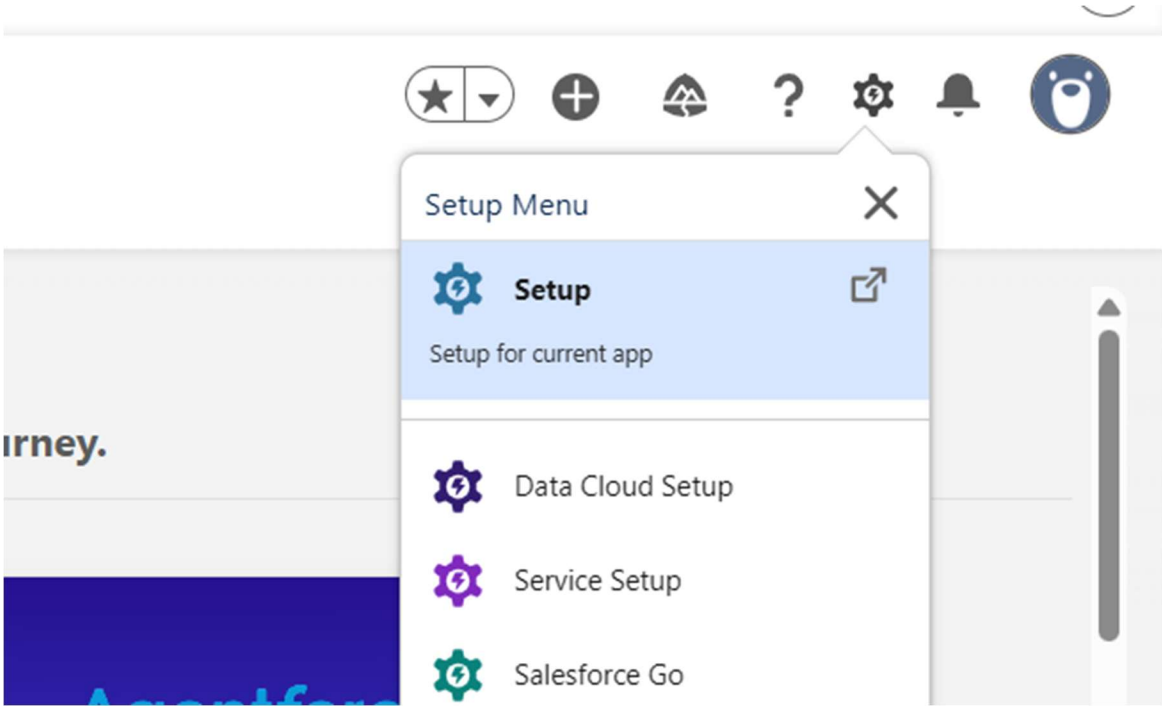
Understanding Tabs

Tabs let you access your objects easily—like browser tabs but for Salesforce.

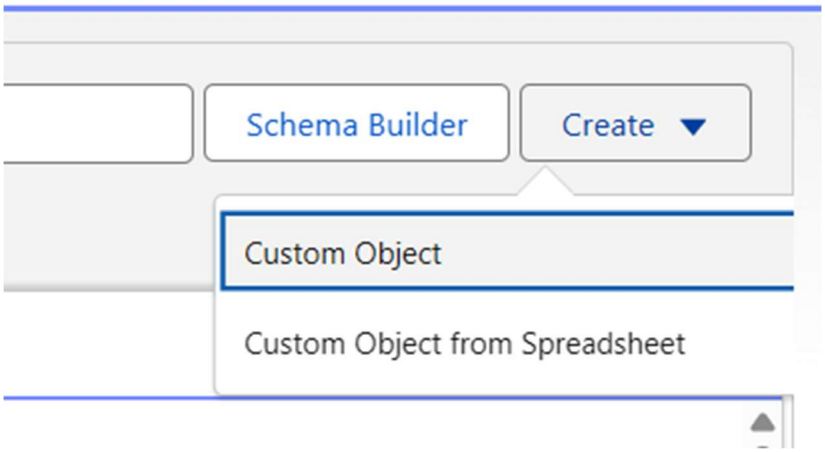
Activity 1: Custom Tabs Creation

For Player Object:

1. Go to Setup



2. Type "Tabs" in Quick Find
3. Under "Custom Object Tabs", click "New"



4. Select "Player__c" from the object list
5. Choose a tab style icon (pick any cricket-related icon!)
6. Click "Next"
7. Add to profiles (keep default)

8. Click "Next"
9. Uncheck "Include Tab" under Custom Apps
10. Make sure "Append tab to users existing personal customizations" is checked
11. Click "Save"

For Match Performance Object:

Repeat the same steps for Match Performance__c object. Pick a different icon!

MILESTONE 4: The Lightning App

Activity 1: Create a Lightning App

A Lightning App is your custom workspace that holds all your tabs!

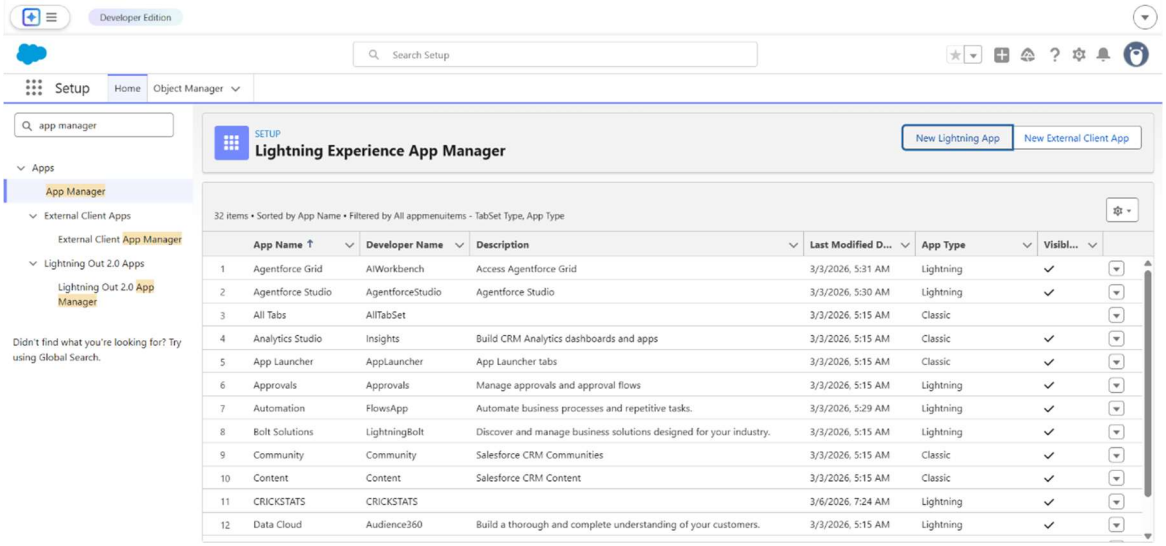
Steps:

1. Go to Setup
2. Search "App Manager" in Quick Find
3. Click "New Lightning App"
4. App Details:
    - App Name: CRICKSTATS
    - Developer Name: CRICKSTATS (auto-fills)
    - Add image: Optional (cricket logo if you want)
    - Primary color: Pick any color or leave default
5. Click "Next"
6. App Options:
    - Navigation style: Standard navigation
    - Setup experience: Setup Supporters
    - Form factors: Desktop and phone ✓
7. Click "Next"
8. Utility Bar: Leave default → Click "Next"
9. Navigation Items (select these tabs):
    - Players
    - Match Performances
    - Reports

- Dashboards

10. Click "Next"
11. User Profiles: Select "System Administrator"
12. Click "Save & Finish"

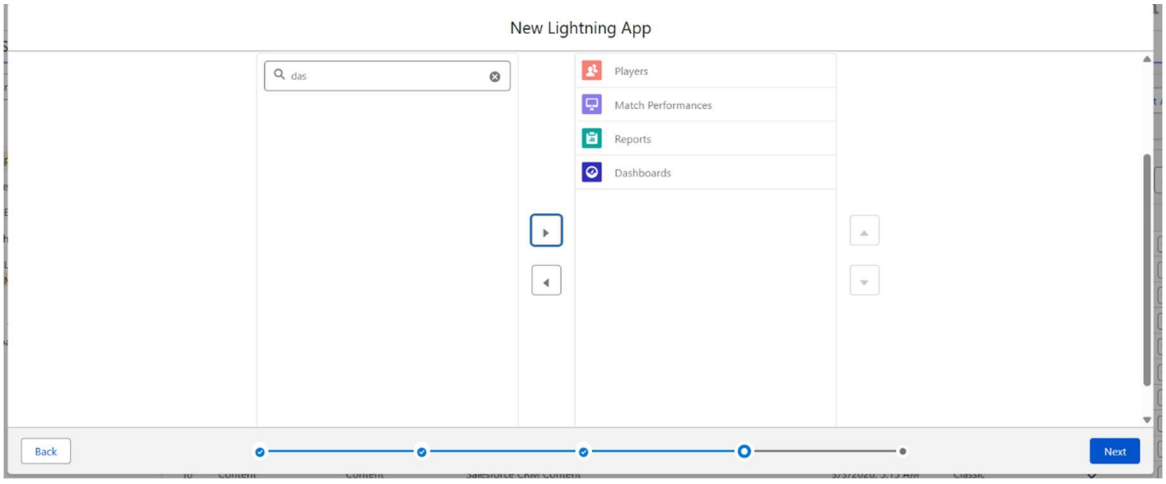
[SCREENSHOT : Lightning App Creation]



To access your app:

1. Click the App Launcher (9-dot waffle icon, top left)
2. Find "CRICKSTATS"
3. Click it—you'll see all your tabs!

[SCREENSHOT : CRICKSTATS App Home with All Tabs]



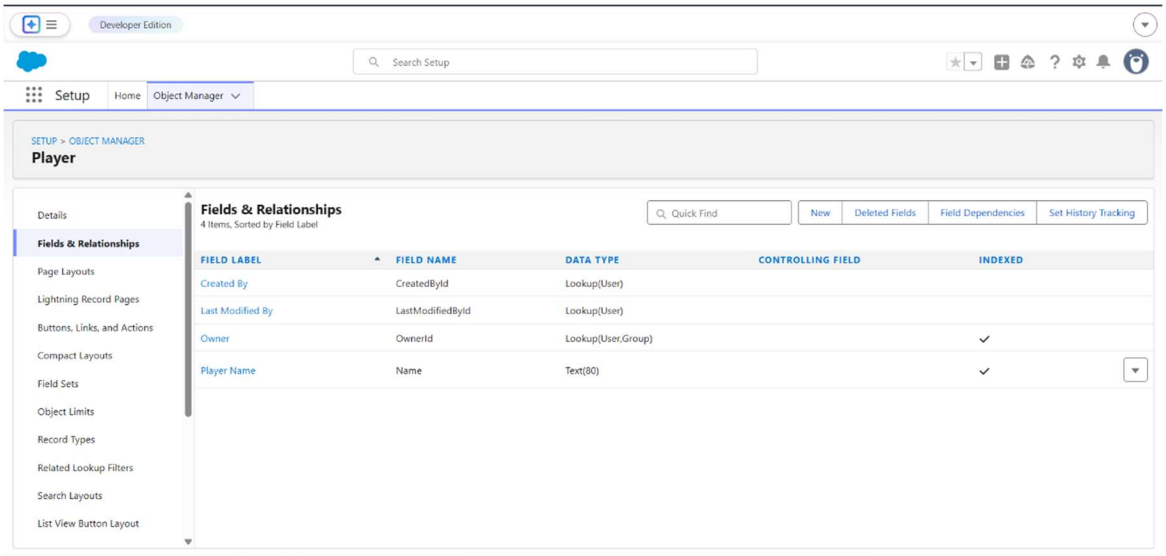
MILESTONE 5: Fields & Relationships

Activity 1: Create Custom Fields for Player Object

Now let's add fields to store player information.

Steps:

1. Go to Setup → Object Manager
2. Search for "Player" object → Click it
3. Click "Fields & Relationships" → "New"



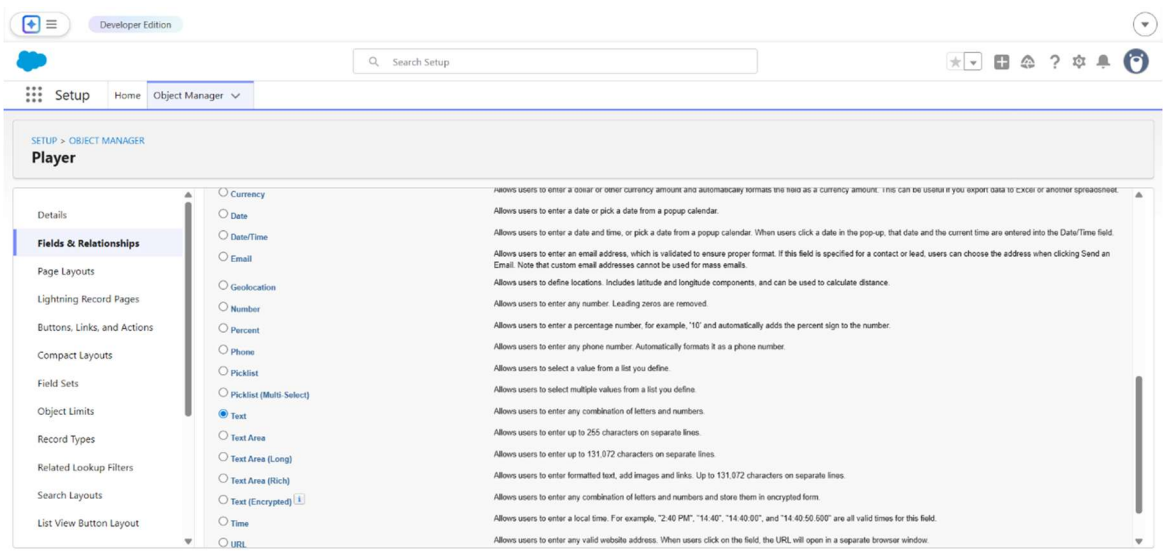
Create these fields one by one:

Field Label	Data Type	Details
Player Name	Text	Length: 50 (already exists as standard)
Player Number	Text	Length: 3 (e.g., "007", "099")
Profile Picture	Rich Text Area	For photo URL
Date of Birth	Date	Player's birth date

Role	Picklist	Batsman, Bowler, Batsman, Wicket Keeper, All Rounder
Batting Style	Picklist	Right-Handed, Left-Handed
Bowling Style	Picklist	Right-Arm Fast, Left-Arm Fast, Right-Arm Medium, Left-Arm Medium, Right-Arm Spin, Left-Arm Spin
Nationality	Picklist	Australia, Bangladesh, England, India, New Zealand, Pakistan, South Africa, Sri Lanka, West Indies, Afghanistan, Zimbabwe, Ireland, Scotland, Netherlands, UAE, Nepal, USA
Team	Picklist	CSK, DC, KKR, MI, PBKS, RR, RCB, SRH, GT, LSG

Table 5: Player object fields (Part 1)

[SCREENSHOT : Creating Player Number Field]



Special Field - AVG Score (Formula):

This automatically calculates batting average!

- Data Type: Formula
- Field Label: AVG Score
- Formula Return: Number
- Formula: IF(Matches_Played__c > 0, Total_Runs__c / Matches_Played__c, 0)

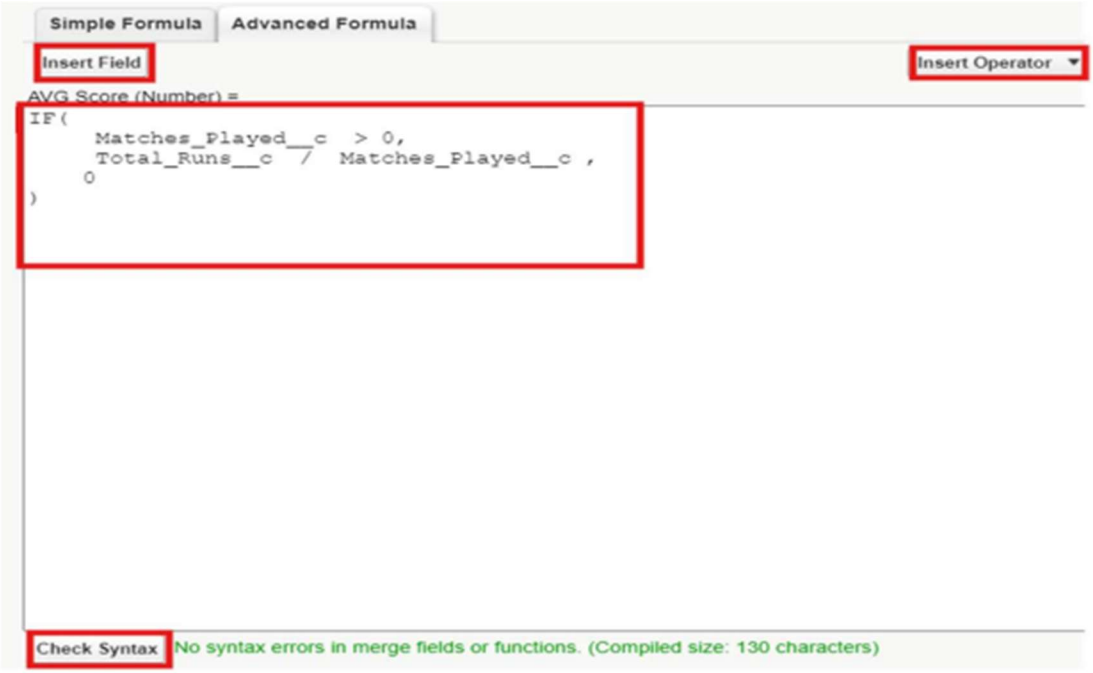
What this does: Divides total runs by matches played to get batting average. If no matches played, shows 0.

Lookup Field - Match Performance:

- Data Type: Lookup Relationship
- Related To: Match Performance
- Field Label: Match Performance

This links player to their match records.

[SCREENSHOT : Player Fields & Relationships List]



Activity 2: Create Custom Fields for Match Performance Object

Now add fields to store match statistics.

Steps:

1. Go to Setup → Object Manager → Match Performance
2. Click "Fields & Relationships" → "New"

Create these fields:

Field Label	Data Type	Details
Match Performance Name	Text	Length: 80 (already exists)
Player of the Match	Checkbox	Was player awarded POM?
Match Result	Picklist	WON, LOST, DRAW
Runs Scored	Number	Batting runs

Balls Faced	Number	Balls faced while batting
Fours	Number	Number of 4s hit
Sixes	Number	Number of 6s hit
Overs Bowled	Number	Overs bowled (e.g., 4.2)
Wickets Taken	Number	Bowling wickets
Runs Conceded	Number	Runs given while bowling
Maiden Overs	Number	Maiden overs bowled
Stumpings	Number	Wicket-keeping stumpings
Catches	Number	Catches taken

Table 6: Match Performance object fields

Master-Detail Relationship Field:

This is the MOST IMPORTANT relationship!

- Data Type: Master-Detail Relationship
- Related To: Player
- Field Label: Player

What this does: Links each match performance to a specific player. When you delete a player, all their match performances are deleted too (that's what Master-Detail means).

MILESTONE 6: Roll-up Summary

Understanding Roll-up Summary

Roll-up summaries are MAGIC! They automatically count, sum, min, or max values from child records (Match Performances) and show totals on the parent record (Player).

For example: Every time you add a match performance with runs scored, the Player's Total Runs field automatically updates!

Note: Roll-up Summary can ONLY be created on the parent object in a Master-Detail relationship.

Activity 1: Create Roll-up Summary Fields for Player Object

Steps:

1. Go to Setup → Object Manager → Player
2. Click "Fields & Relationships" → "New"
3. Choose "Roll-up Summary" → Click "Next"

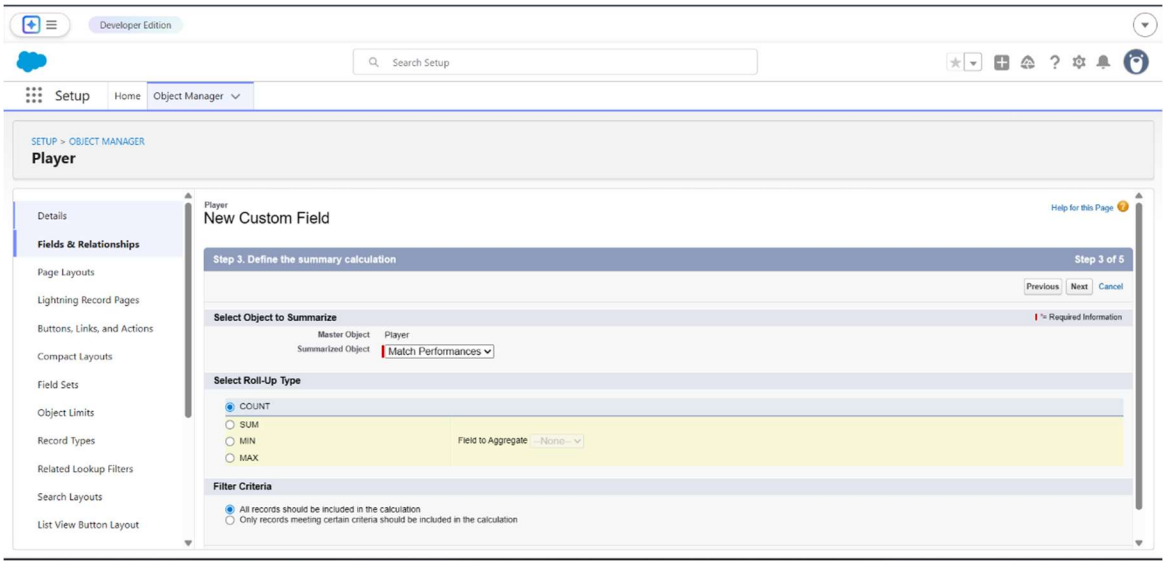
Create these roll-up fields one by one:

1. Matches Played (COUNT):

- Field Label: Matches Played
- Summarized Object: Match Performances
- Roll-up Type: COUNT
- Filter Criteria: All records should be included

Counts how many match performance records exist for this player.

[SCREENSHOT : Creating Matches Played Roll-up Summary]

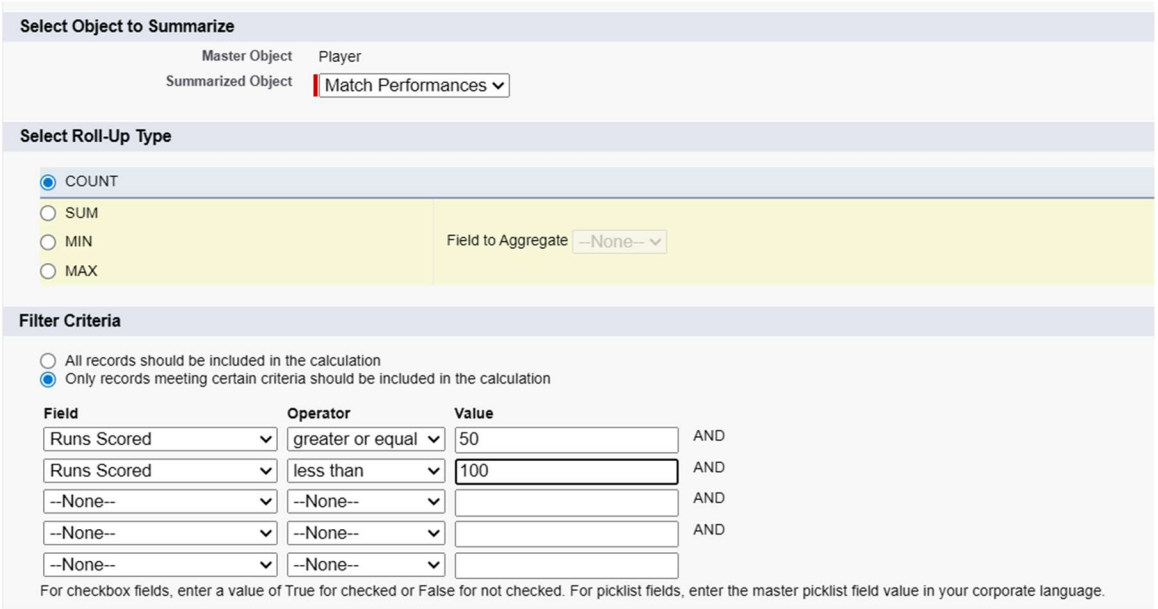


2. Number of 50s (COUNT with filter):

- Field Label: Number of 50s
- Summarized Object: Match Performances
- Roll-up Type: COUNT
- Filter Criteria: Only records meeting certain criteria
- Criteria: Runs_Scored__c >= 50 AND Runs_Scored__c < 100

Counts matches where player scored 50-99 runs.

[SCREENSHOT : Number of 50s Roll-up with Filter Criteria]



3. Number of 100s (COUNT with filter):

- Field Label: Number of 100s
- Summarized Object: Match Performances
- Roll-up Type: COUNT
- Filter Criteria: Only records meeting certain criteria
- Criteria: Runs_Scored__c >= 100

Counts matches where player scored century (100+ runs).

4. Total 4s (SUM):

- Field Label: Total 4s
- Summarized Object: Match Performances
- Roll-up Type: SUM
- Field to Aggregate: Fours
- Filter Criteria: All records

Adds up all fours hit across all matches.

5. Total 6s (SUM):

- Field Label: Total 6s
- Summarized Object: Match Performances
- Roll-up Type: SUM
- Field to Aggregate: Sixes

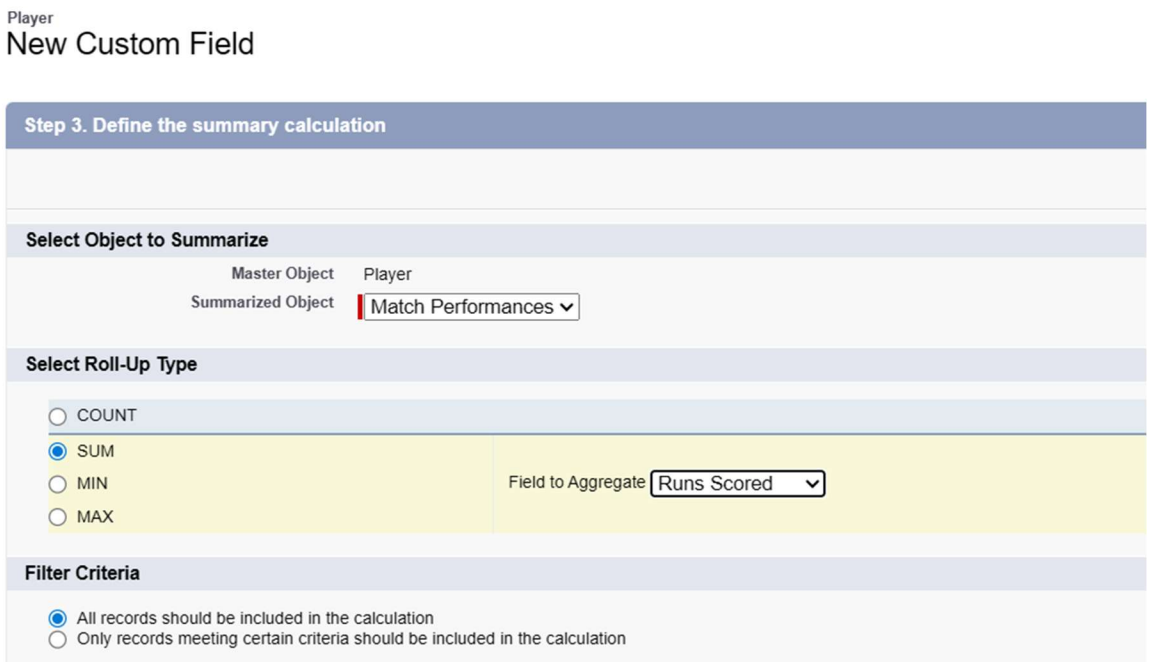
- Filter Criteria: All records

Adds up all sixes hit across all matches.

6. Total Runs (SUM):

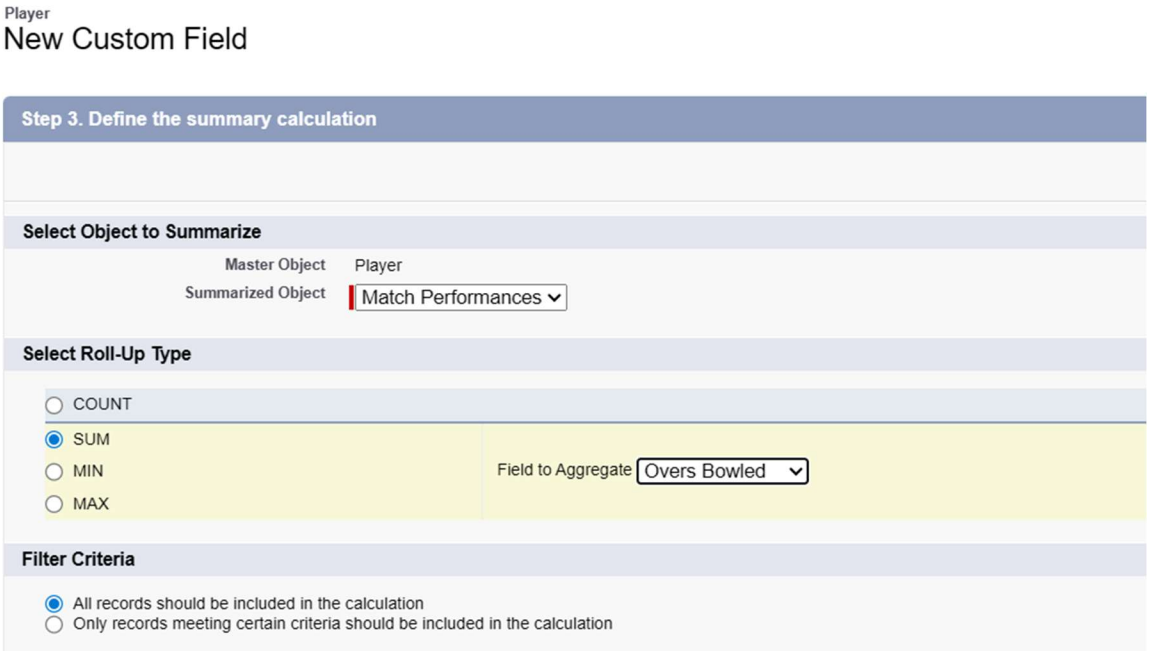
- Field Label: Total Runs
- Summarized Object: Match Performances
- Roll-up Type: SUM
- Field to Aggregate: Runs Scored
- Filter Criteria: All records

Important: This field is used by the AVG Score formula!



7. Total Overs (SUM):

- Field Label: Total Overs
- Summarized Object: Match Performances
- Roll-up Type: SUM
- Field to Aggregate: Overs Bowled
- Filter Criteria: All records



8. Total Wickets (SUM):

- Field Label: Total Wickets
- Summarized Object: Match Performances
- Roll-up Type: SUM
- Field to Aggregate: Wickets Taken
- Filter Criteria: All records

9. Total Stumpings (SUM):

- Field Label: Total Stumpings
- Summarized Object: Match Performances
- Roll-up Type: SUM
- Field to Aggregate: Stumpings
- Filter Criteria: All records

10. Total Catches (SUM):

- Field Label: Total Catches
- Summarized Object: Match Performances
- Roll-up Type: SUM
- Field to Aggregate: Catches
- Filter Criteria: All records

What happens now:

Every time you create a Match Performance record:

- Player's Matches Played automatically increases by 1
- Player's Total Runs increases by the runs scored
- Player's Total 4s, 6s, Wickets, Catches all update
- Player's AVG Score recalculates automatically!

It's automatic magic!

MILESTONE 7: Flows

Understanding Flows

Flows are workflows that automate tasks. For CRICKSTATS, we'll create a Screen Flow that lets users select a team and see all players from that team with their statistics.

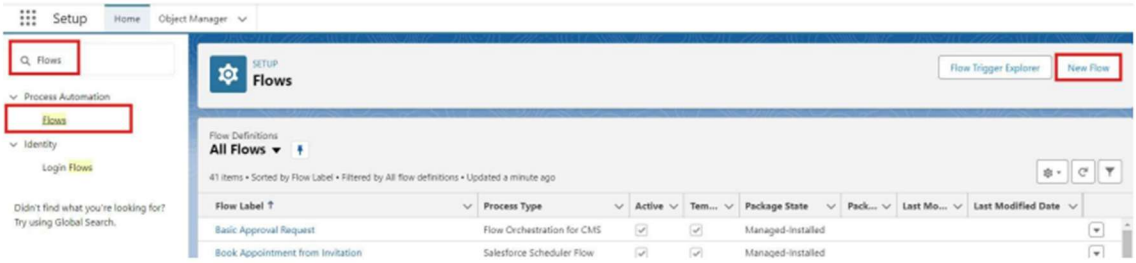
Activity 1: Screen Flow

This creates an interactive screen where users can filter players by team!

Steps:

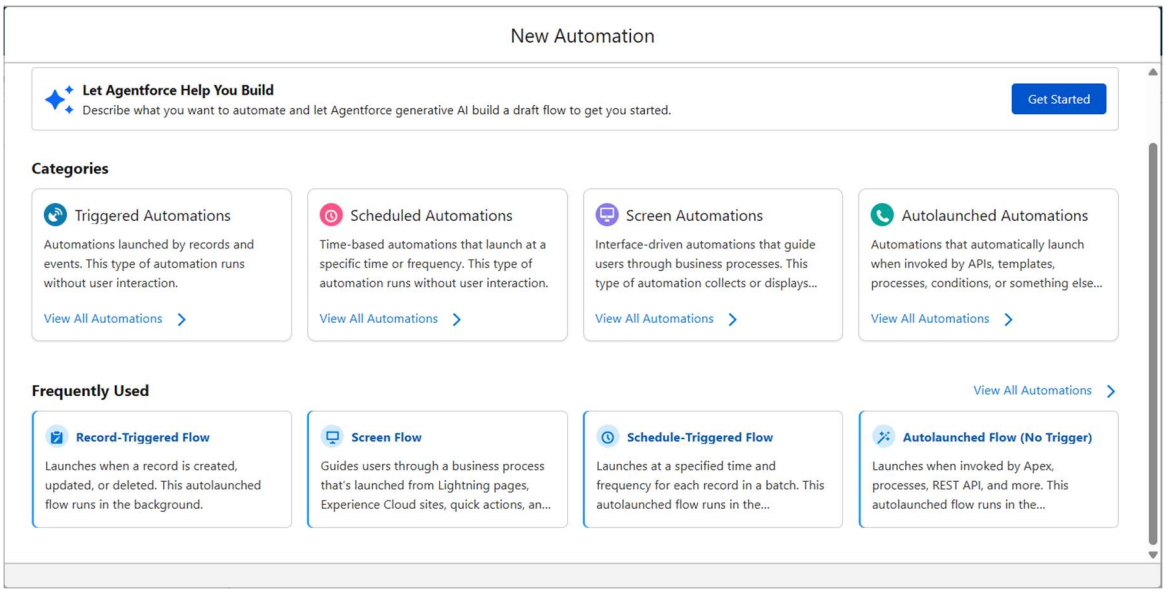
1. Go to Setup
2. Type "Flow" in Quick Find → Click "Flows"
3. Click "New Flow"
4. Select "Screen Flow" → Click "Create"

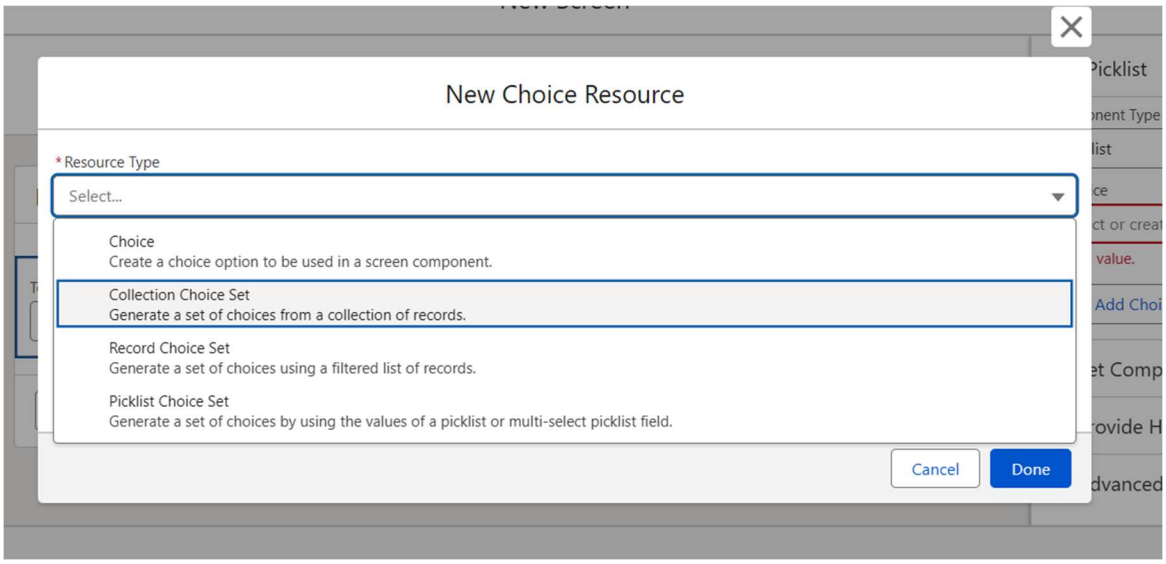
[SCREENSHOT : Flow Builder - New Screen Flow]



Screen 1: Team Selection

1. Click "+" icon on the flow builder
2. Select "Screen" component
3. Label: Players screen

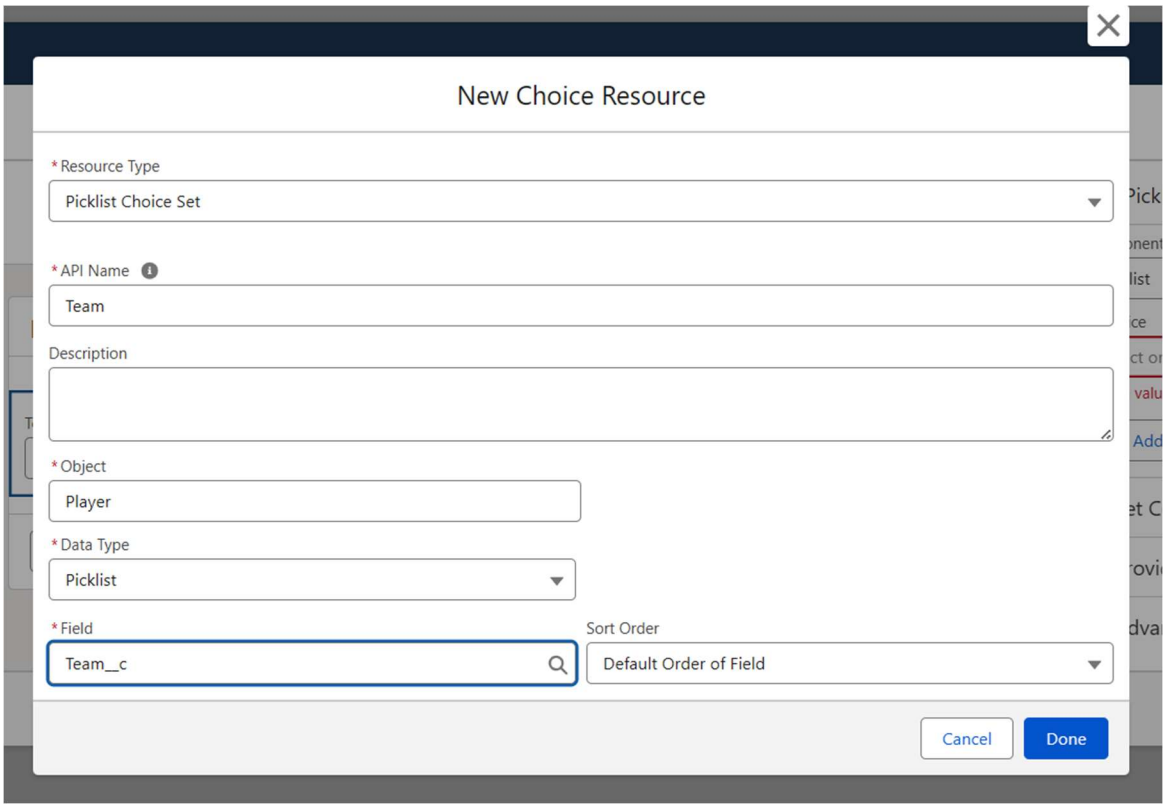




4. API Name: Playersscreen
5. From left panel, drag "Picklist" component onto the screen
6. Configure Picklist:
    - Label: Team
    - API Name: Team
    - Component Type: Picklist
    - Data Type: Text
    - Click "New Choice Resources" → "Picklist Choice Set"
    - API Name: Team

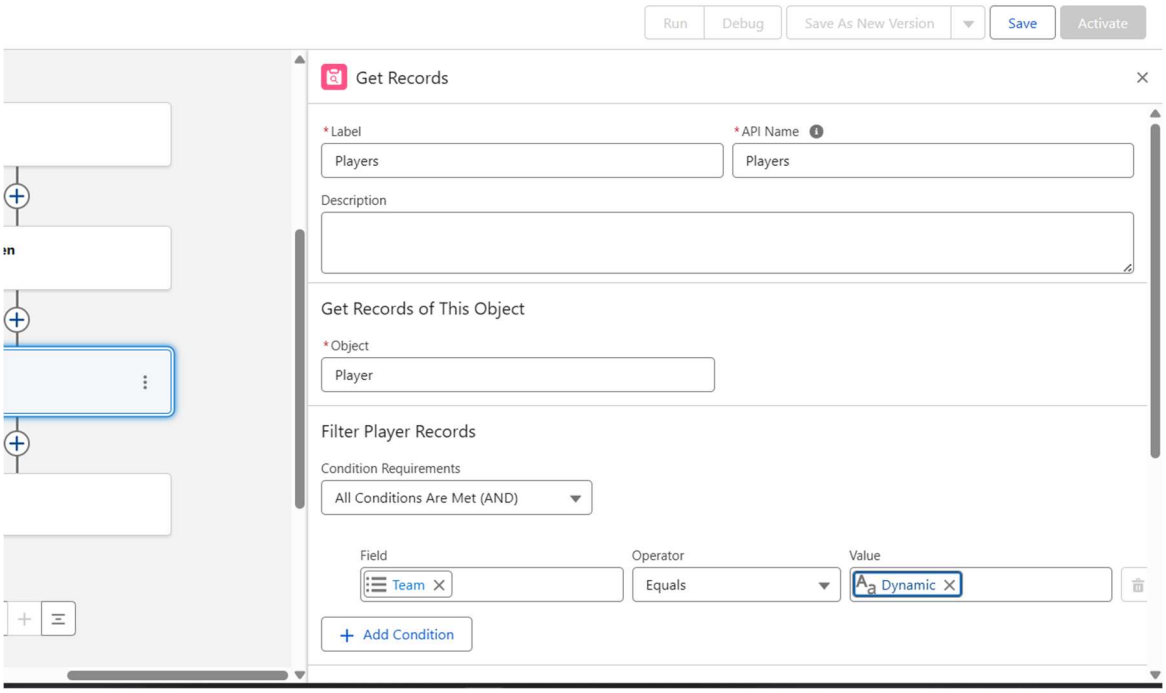
- Object: Player
- Data Type: Picklist
- Field: Team
- Click "Done"

7. Click "Done"

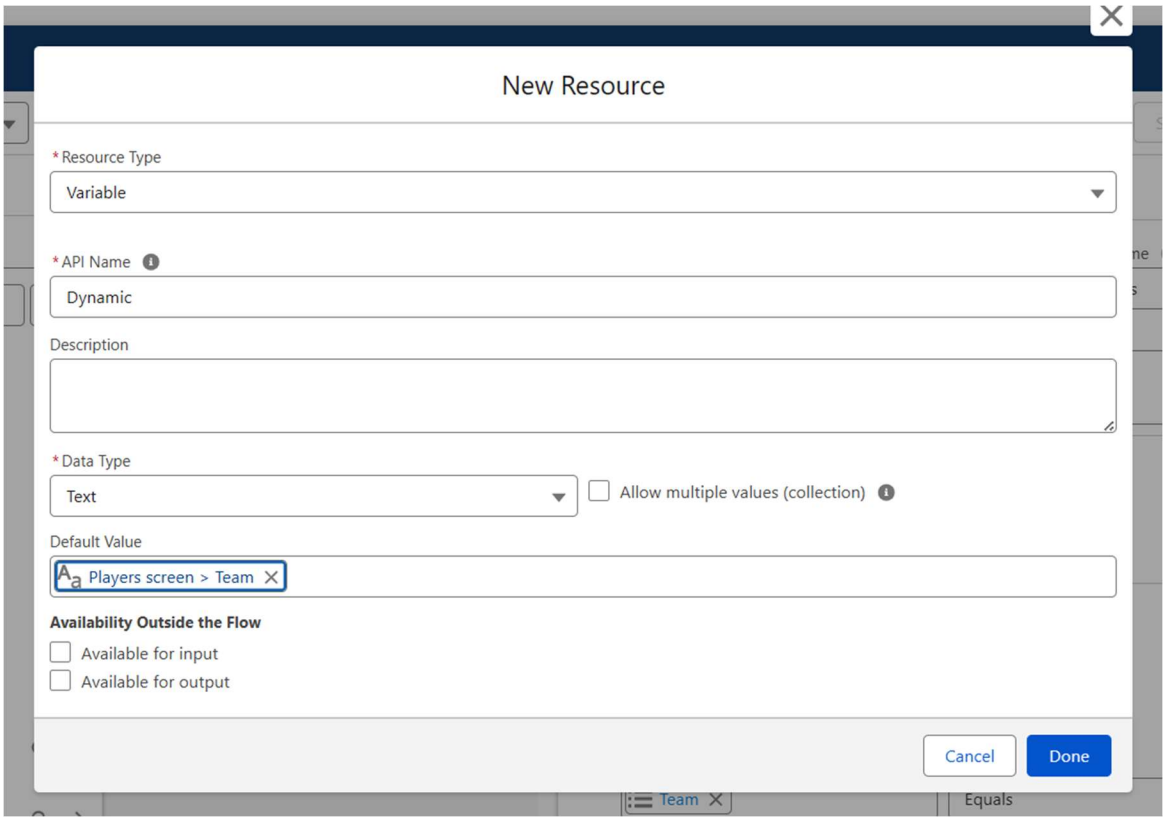


Get Records: Fetch Players by Team

1. Click "+" icon
2. Select "Get Records" component
3. Label: Players
4. API Name: Players
5. Object: Player



6. Filter player records:
    - Field: Team
    - Operator: Equals
    - Value: Click "New Resource" → "Variable"
    - Resource Type: Variable
    - API Name: Dynamic
    - Data Type: Text
    - Default Value: {!Team} (from screen component Team Picklist)
7. Click "Done"



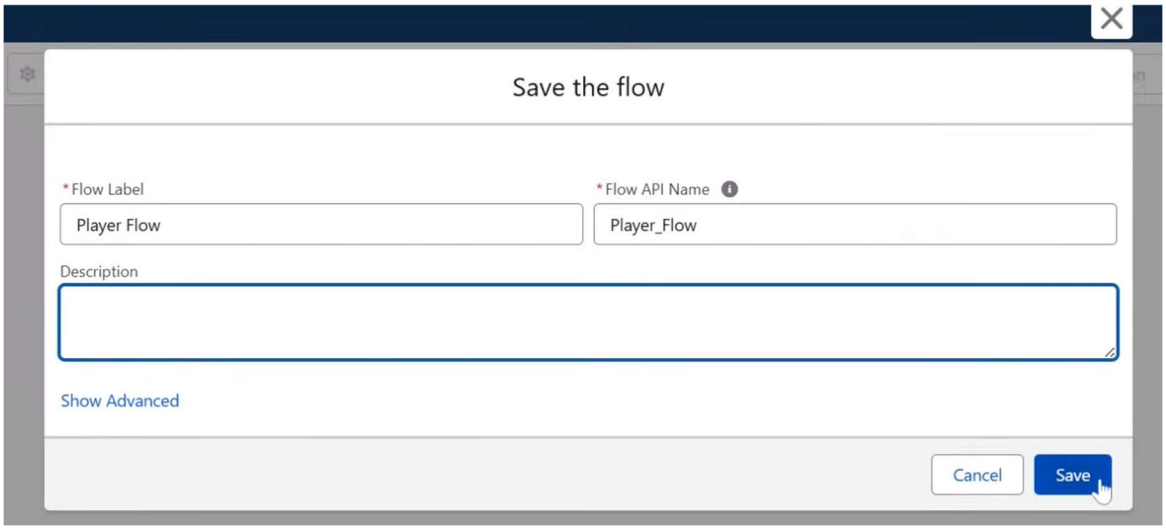
Screen 2: Display Player Info in Data Table

1. Click "+" icon
2. Select "Screen" component
3. Label: Playerinfo
4. API Name: Playerinfo
5. From left panel, drag "Data Table" component onto screen
6. Configure Data Table:
    - API Name: Playerinfotable
    - Label: Data Table
    - Configure Data Source:
        - Source: Collection
        - Collection: Players (from Get Records)
        - Object: Player__c
    - Configure Columns (Add these columns):
        - Profile Picture

    - Player Name
    - Total Runs
    - AVG Score
    - Batting Style
    - Bowling Style
    - Matches Played
    - Number of 50s
    - Number of 100s
7. Click "Done"

Save and Activate:

1. Click "Save"
2. Flow Label: Player Flow
3. Flow API Name: PlayerFlow
4. Click "Save"
5. Click "Activate"



Activity 2: Add Flow to Home Page

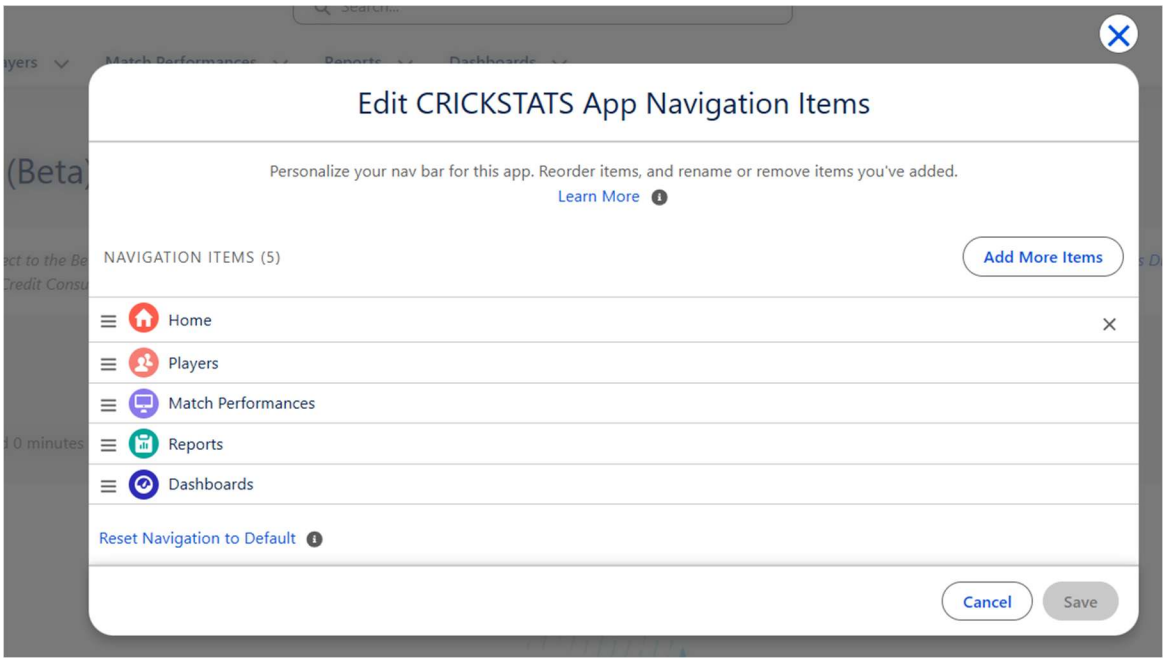
Now let's add this flow to your CRICKSTATS home page so users can access it easily!

Steps:

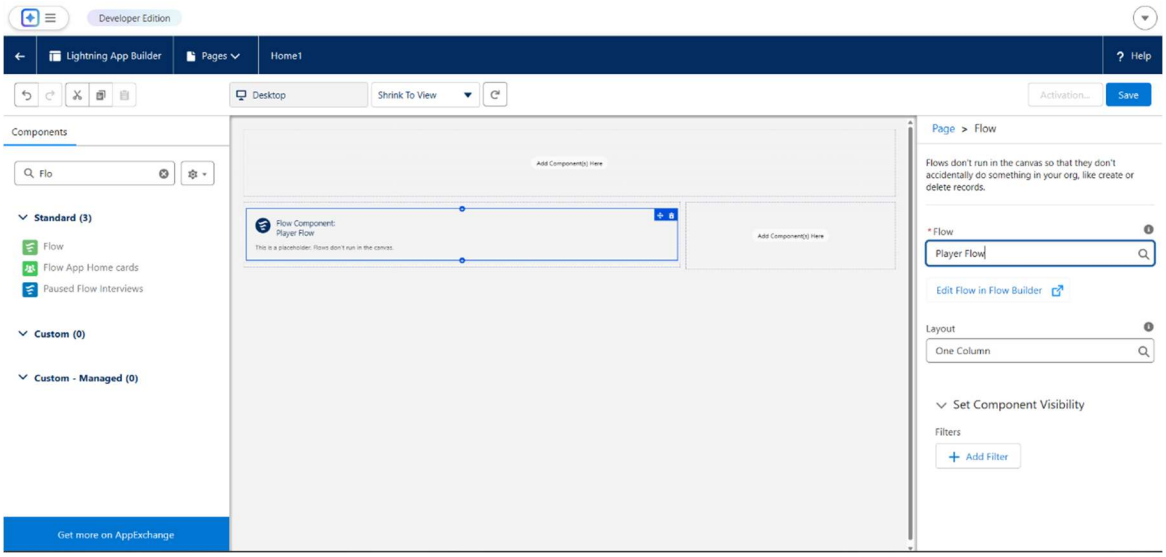
1. In your CRICKSTATS app, click the pencil icon (Personalize your nav bar) on top right of Navigation Bar
2. Add "Home" page to navigation

3. Click "Save"
4. Go to Home page
5. Click gear icon → "Edit Page"
6. Remove all default components (drag them off)
7. From left panel under "Standard" components, drag "Flow" component onto the page
8. Configure Flow:
    - Flow: Players Flow
9. Click "Save"
10. Click "Activation"
11. Click "App Default"
12. Click "Assign to Apps"
13. Select "CRICKSTATS" app
14. Click "Next" → "Save"

[SCREENSHOT : Lightning Page Editor with Flow Component]

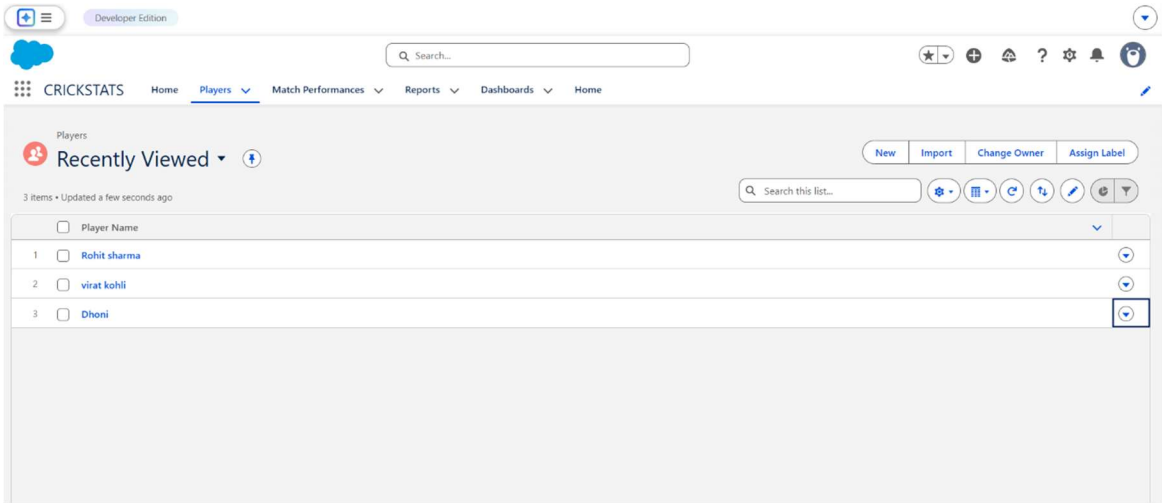


[SCREENSHOT : Flow Component Configuration on Home Page]



Test it:

1. Go to CRICKSTATS app → Home page
2. You'll see a dropdown with IPL teams
3. Select a team (e.g., "CSK")
4. Click "Next"
5. See all CSK players with their statistics in a table!



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7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce Lightning user interface to verify the correct functioning of all implemented features in the CRICKSTATS system.

What we tested:

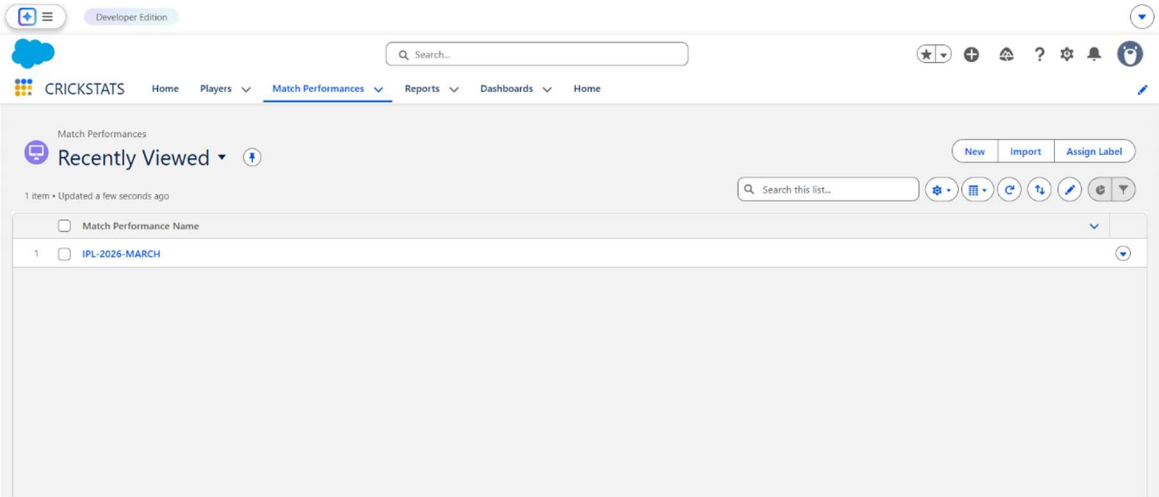
- **Record creation** - Created Player and Match Performance records
- **Validation checks** - Tested mandatory fields, score limits, input constraints
- **Record types** - Verified behavior for Domestic and International match categories (if applicable)
- **Flow execution** - Tested automatic updating of player statistics and data consistency
- **Roll-up summary fields** - Verified accurate calculation of total runs, wickets, averages, and performance metrics
- **System performance** - Monitored response time during data entry, report generation, and dashboard loading
- **Multiple user interactions** - Tested stable performance without delays
- **Reports and dashboards** - Confirmed accurate data visualization, real-time updates, and reliable performance tracking

Testing Steps:

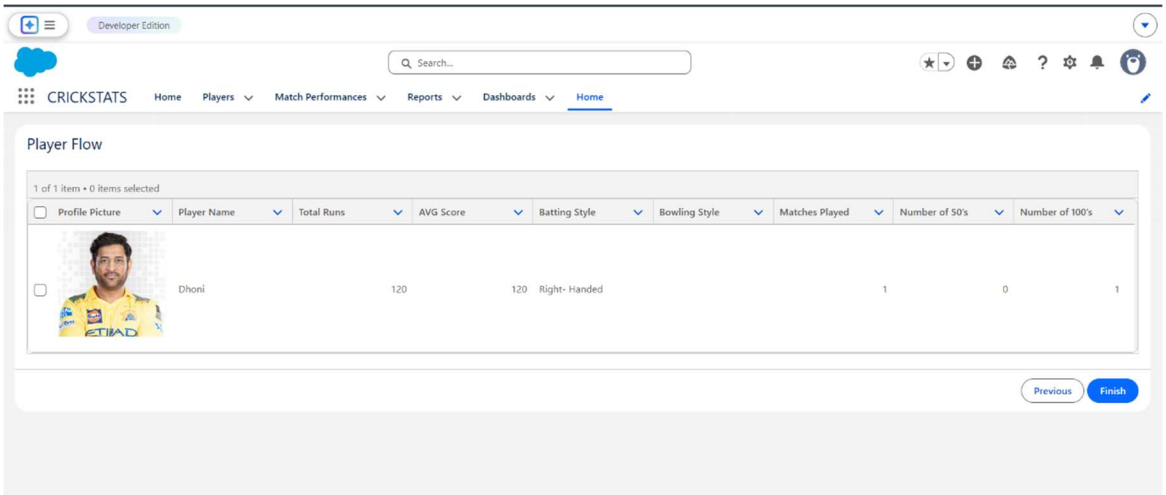
1. **Enter some test data:**
    - Create 5-10 player records with different teams
    - For each player, create 2-3 match performance records
    - Enter realistic statistics (runs, wickets, catches)
2. **Verify roll-up summaries:**
    - Check that Total Runs, Matches Played update automatically
    - Verify AVG Score formula calculates correctly
    - Confirm 50s and 100s count correctly
3. **Test the flow:**
    - Go to Home page
    - Select one team from the picklist
    - Click "Next"
    - Verify that only players from selected team appear
    - Check that all statistics display correctly in the data table
4. **Generate reports:**
    - Create report showing players by team

    - Create report showing top run-scorers
    - Create report showing top wicket-takers
- *Add screenshot of a Player record showing all fields filled with realistic data and roll-up summary fields populated*

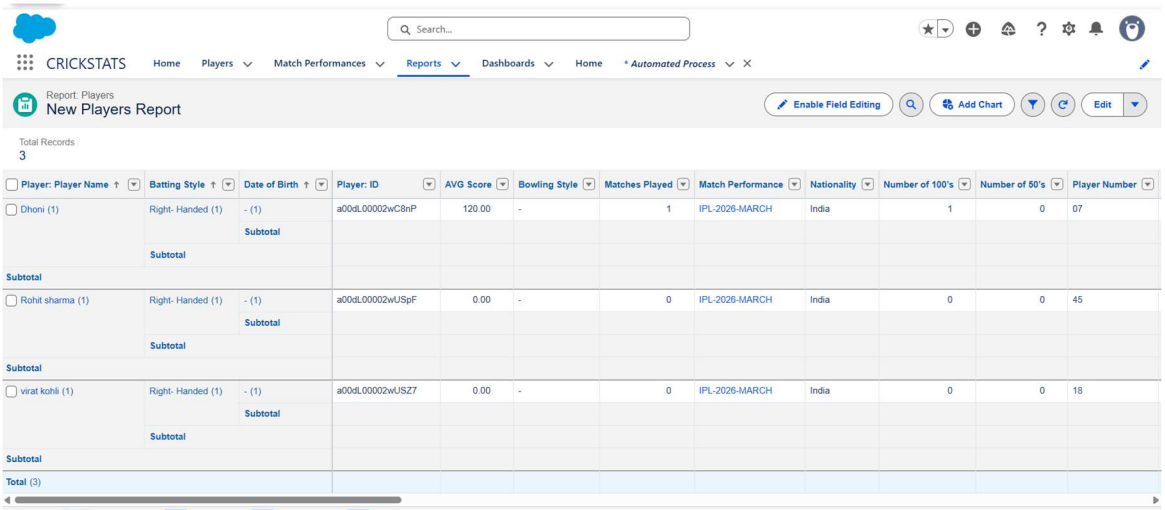
[SCREENSHOT : Sample Match Performance Record]



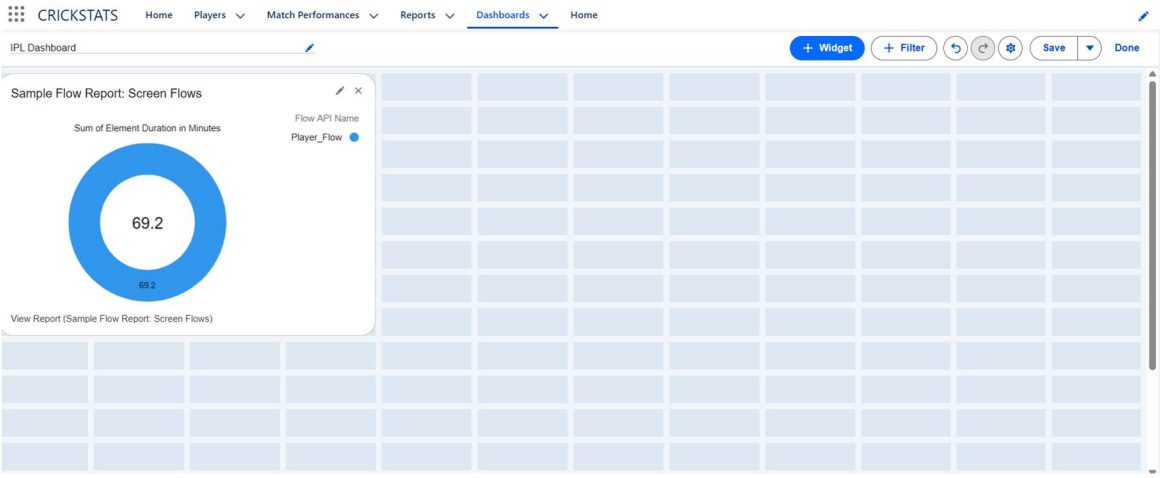
[SCREENSHOT : Sample Player Record with Test Data]



[SCREENSHOT : Report - Top Run Scorers]



[SCREENSHOT : Dashboard - Performance Overview]



These testing activities ensured that the CRICKSTATS platform met both functional and performance requirements and was ready for deployment.

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8. ADVANTAGES & DISADVANTAGES

Advantages

- **Centralized system** - All player data in one place, no more scattered files
- **Real-time analytics** - See player statistics instantly
- **Secure platform** - Role-based access controls protect sensitive data
- **Scalable design** - Can expand to multiple teams and leagues
- **Reduced manual work** - Roll-up summaries calculate stats automatically
- **Data accuracy** - Validation rules prevent errors

- **Easy reporting** - Generate reports in seconds
- **Cloud-based** - Access from anywhere, no installation needed
- **Master-Detail relationships** - Strong data integrity between players and matches

Disadvantages

- **Requires Salesforce license** - Need subscription after developer org
- **Internet dependency** - Requires internet connection to access
- **Limited offline access** - Can't work offline
- **Learning curve** - Users need to learn Salesforce interface
- **Customization limits** - Some complex features may need Apex code

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9. CONCLUSION

CRICKSTATS successfully implements a cloud-based cricket performance tracking system using Salesforce. The system automates data management, improves reporting accuracy, and supports informed decision-making.

Key achievements:

- 2 custom objects (Player, Match Performance)
- 25+ custom fields across both objects
- 1 Master-Detail relationship (Match Performance → Player)
- 1 Lookup relationship (Player → Match Performance)
- 10 roll-up summary fields for automatic calculations
- 1 formula field (AVG Score)
- 1 interactive screen flow for team-based player filtering
- Custom Lightning App with tabs
- Reports and dashboards for performance analytics

The project demonstrates practical application of:

- SDLC (Software Development Life Cycle)
- Agile methodology with sprint planning
- CRM-based development using Salesforce
- Object relationships and data modeling
- Automation with flows and roll-up summaries

CRICKSTATS transforms manual cricket statistics management into a scalable, reliable, and user-friendly digital solution!

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10. FUTURE SCOPE

Potential enhancements for the system:

- **Mobile application integration** - Access CRICKSTATS on mobile devices
- **AI-based performance prediction** - Predict player form and match outcomes
- **Video analysis integration** - Link video clips to match performances
- **Player ranking system** - Automatic ranking based on performance metrics
- **Public statistics portal** - Share player stats publicly (fan engagement)
- **Multi-league support** - Expand to IPL, BBL, CPL, international cricket
- **Advanced analytics** - Strike rate, economy rate, bowling average calculations
- **Automated notifications** - Alert coaches when player crosses milestones (50, 100)
- **Integration with live scores** - Auto-populate match data from live score APIs
- **Historical data comparison** - Compare current vs past season performance

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11. APPENDIX

Salesforce Objects

Player Object:

- Custom fields: Player Number, Profile Picture, Date of Birth, Role, Batting Style, Bowling Style, Nationality, Team
- Roll-up fields: Matches Played, Total Runs, Total 4s, Total 6s, Total Overs, Total Wickets, Total Stumpings, Total Catches, Number of 50s, Number of 100s
- Formula fields: AVG Score

Match Performance Object:

- Custom fields: Player of the Match, Match Result, Runs Scored, Balls Faced, Fours, Sixes, Overs Bowled, Wickets Taken, Runs Conceded, Maiden Overs, Stumpings, Catches
- Master-Detail relationship to Player

Flow Configurations

Player Flow:

- Screen 1: Team selection picklist
- Get Records: Fetch players by selected team
- Screen 2: Data table displaying player statistics

Manual Match Statistics

Process for entering match data:

1. Create Player records first
2. After match completion, go to Match Performances tab
3. Click "New"
4. Select Player (lookup)
5. Enter match statistics (runs, wickets, catches, etc.)
6. Save
7. Check Player record—roll-up summaries update automatically!

TEAM CONTRIBUTIONS

Team Member	Contributions
PRITHIVIRAJ L	Developer Account, Fields Creation, Flows, Overall Documentation
ABISHEK D	Custom Objects, Page Layouts, Data Validation
RAJA I	Tabs Creation, Navigation Setup
HARIKARAN E	Lightning App, UI Design, Automation

Table 7: Team member contributions

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[2] Salesforce Trailhead. (2025). *Admin Beginner Trail*. https://trailhead.salesforce.com

[3] Salesforce Help. (2025). *Flow Builder Guide*.
https://help.salesforce.com/s/articleView?id=sf.flow.htm

[4] Salesforce Help. (2025). *Roll-Up Summary Fields*.
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[5] Salesforce Help. (2025). *Master-Detail Relationships*.
https://help.salesforce.com/s/articleView?id=sf.relationships_considerations.htm

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Thank You!

This documentation was prepared by **PRITHIVIRAJ L** and team for the CRICKSTATS Cricket Player Performance Tracking Application at E.S ARTS AND SCIENCE CO-ED COLLEGE.

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